



Kepler Telescope Recovered

NASA has recovered Kepler Space Telescope which slipped into 'emergency mode' when it was out for planet-hunt. <http://dgit.in/KSTnasa>



Solar Impulse 2

Solar Power Airplane- Solar Impulse 2 is on it's round the world tour. <http://dgit.in/Slrmp>

Space Age



One of the LIGO facilities where the experiments were conducted

the experiment. This explains why there are two locations for the experiments; by effectively identifying only the signals that are common to both locations, the experimentalists can filter out the signals produced due to noise at each individual location. That being said, we still require extremely large masses to produce fields of measurable strengths. It took a system as monumentally huge as two revolving black holes, some of the densest and most massive substances in the galaxy, to produce detectable waves. The announcement was made on 11th of February, 2016, and the signals that were observed matched the predictions of general relativity and perfectly complied with the relativistic description of the inward spiral and merger of two black holes. Physics works!

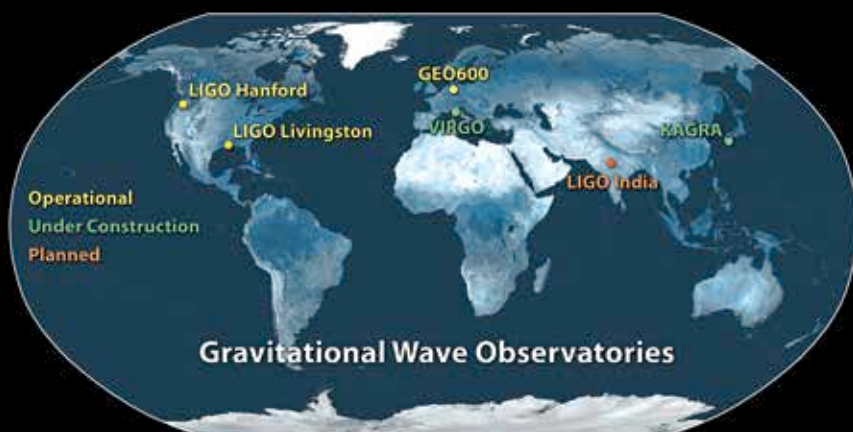
Now that the first observations have been made, it is possible for scientists to observe phenomena that occur at intergalactic distances. Any gravitational wave that is produced is matched to a series of templates that are computationally generated to match relativistic predictions of different events. When a match has been found for a particular signal, the experimentalists and the cosmologists can then theorise how the cosmic event occurred. In a lot of ways, this greatly improves the range of data that is available to scientists, who no longer have to depend solely on the radiation and emission of light spectra to analyse astronomical data.

It is imperative to not only improve the experimental apparatus at the current locations but also build more such interferometers for improved detection and noise

filtration. In a landmark project, the Union Cabinet of the Indian Government has nearly finalised a proposal to construct a similar experimental facility. The project is a joint venture by the LIGO Laboratory and three leading institutions in the Indian Initiatives in Gravitational-Wave Observations (IndIGO) consortium: Institute of Plasma Research (IPR) Gandhinagar, Inter University Centre for Astronomy and Astrophysics (IUCAA), Pune and Raja Ramanna Centre for Advanced Technology (RRCAT), Indore. LIGO USA labs have agreed to donate most interferometer components but the burden of building the entire infrastructure, facilities and interferometer vacuum housing, and then assembling, commissioning and operating the detector, lies in the hands of the Indian scientific community. According to Professor C Unnikrishnan of the Tata Institute of Fundamental Research, who is a key member of the LIGO India project, the project 'give(s) both scientists and industries technological goals that go beyond what is available or possible in the country today, leading to enhancement of technological capabilities'. The impact on higher education and research in Physics and

Astronomy is also expected to be monumental, due to what Prof Unnikrishnan termed an 'unprecedented potential for human resource development in several areas of physics, technology and engineering'. Professor Bala Iyer of the Raman Research Institute, also a key member of the project, hailed the proposal as a 'historical epoch' and believes that LIGO-India will be a 'critical element of the global network needed to inaugurate the new GW astronomy with the potential to impact R&D in the many technologies involved and in turn impact high precision experiments in many areas'.

The next few years will not only see a new experimental setup in India, it will also see many bright minds step up to take the mantle that has been created by this proposal. Gravitational wave observations hold the answers to many deep questions that trouble physics. The stimulus now exists for the next generation of physicists to step up and continue the noble quest of science to unravel the mysteries of the universe. 



LIGO Facility planned for India in the coming years